Round 6: early-stopping F4 and the trace coset on every Koblitz curve

research/isogeny_conductor_gap_round6_20261018, 2026-10-18. E8/E9a: 6,000 descent systems on 5 Koblitz curves. E9b: one audited ecbench session (168 records, 12 replays reproduced).

Findings

1. **Lead 1 does not carry over to the binary descent.** A solvable system costs 1.0 to 2.3 times an unsolvable one. Stopping F4 once its degree-1 elements leave 8 unknowns saves up to 43 %, but only where most systems are solvable; it costs up to 37 % more where few are. Solutions are identical on all 6,000 systems.
2. **The trace coset's yield gain is $1/f_2$** (7 of 9 rows). f_2 is 0.45 at F_2^{19}, which refutes round 5's explanation.
3. **The affine coset extends the yield gain to a = 1** (1.6 to 3.1 times). F4 work per system rises 1.1 to 45 times; the large factors come where the solving degree goes from 3 to 4, on a = 0 too.
4. **Whole method on a = 1:** coset MITM beats default MITM across the band at n = 19 and 23. Best cell: 12.7 times rho. The study's best stays 7.29 times rho.

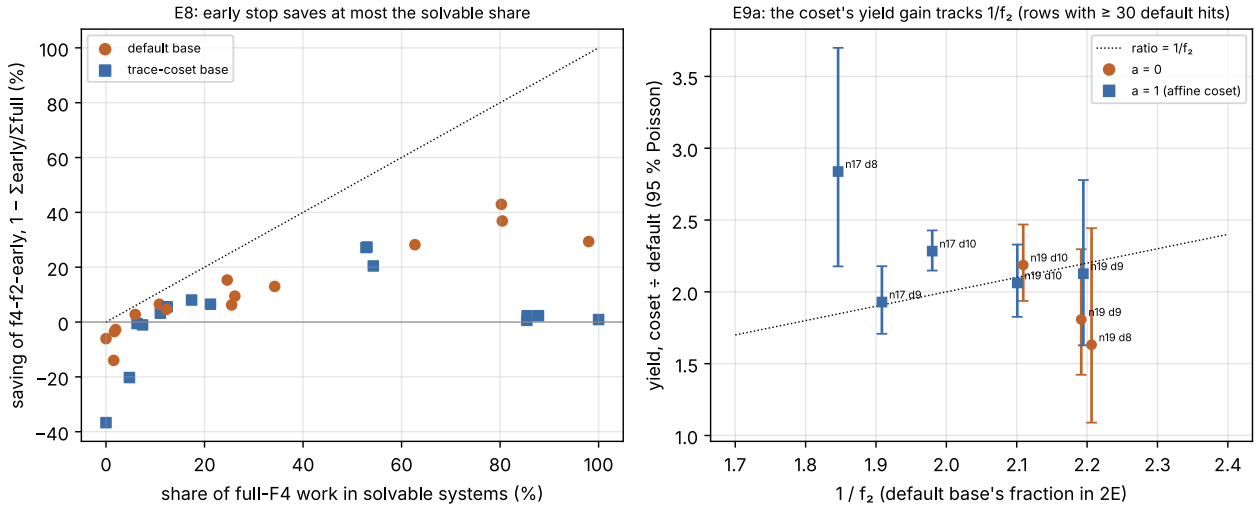


Figure 1: E8 saving against solvable share; E9a yield ratio against $1/f_2$.

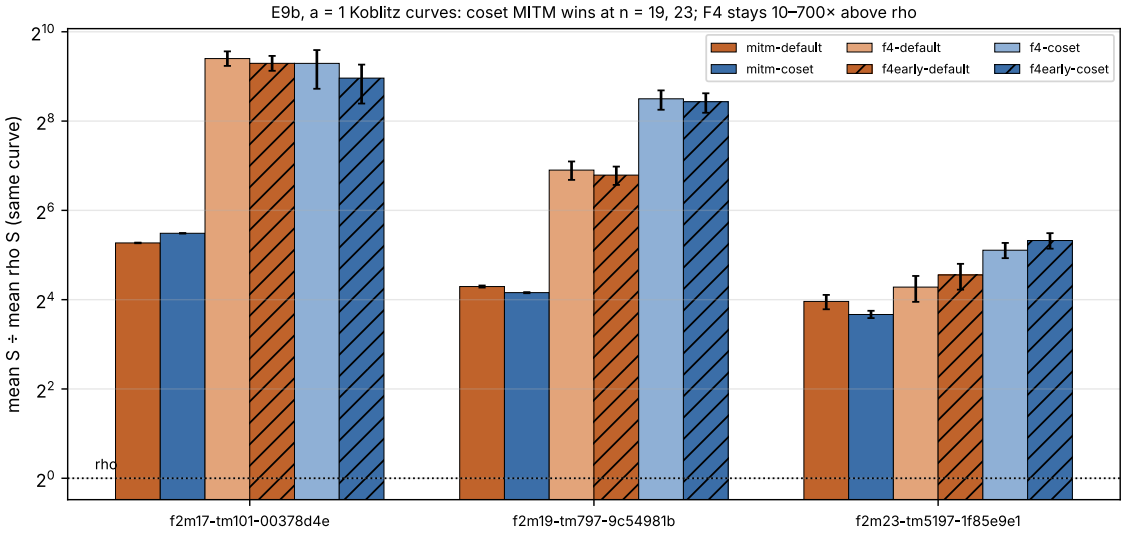


Figure 2: E9b: S over rho per arm per curve (a = 1 Koblitz).

Leads

1. Choose the early-stop threshold from the expected solvable share, or use free = 2 to 4.
2. A 2E base inside the default's span, without z^d, may keep the solving degree at 3.

3. Round 5′s F_2^{19} yield of 1.48 times is unexplained (per system: 1.81). Check whether the ecbench walk targets are evenly spread.
4. Binary MITM stays 13 to 45 times above rho; no lever changes the exponent.